A strict pointwise tensor-train approximation bound by finite singular-subspace tie handling

Matthew J. Colbrook*

11 September 2026

Recovered qualified manuscript for TR-04. Not a uniform-factor improvement.

Authorship recorded at the submitter's request. This recovered draft was generated with AI assistance. The complete original source passed independent agent review against the exact repository target. This is not external human peer review or formal certification; no priority claim is made. Original provenance, full-source review hashes and diagnostic results accompany this manuscript in the submission record.

Abstract

For prescribed tensor-train rank bounds, a deterministic algorithm in the exact arithmetic/SVD model returns an approximation with squared error strictly smaller than $(d-1)$ times the optimal squared error whenever that optimum is positive, and reconstructs exactly at zero optimum. It performs at most n_1 tensor-train SVD completions. The only modification to the standard construction is to test a finite collection of equally optimal first-cut singular subspaces when a positive singular value straddles the truncation boundary. This addresses the strict pointwise inequality displayed in TR-04. It does not produce a uniform constant $c < d-1$: an explicit fixed-format family shows that its error ratios can approach $d-1$ already when $d = 3$.

1 The target and the distinction between two guarantees

Let $d \geq 3$, let $A \in \mathbb{R}^{n_1 \times \cdots \times n_d}$ be a dense tensor, and let $r_1, \ldots, r_{d-1}$ be positive integers. Denote by S_r the set of tensors whose unfolding at every cut $1, \ldots, j \mid j+1, \ldots, d$ has rank at most r_j. Write

$$E_* = E_*(A, r) = \min_{Y \in S_r} \|A - Y\|_F^2.$$

This minimum exists: S_r is a closed intersection of determinantal sets, and a minimizing sequence may be confined to a bounded ball. The displayed strict inequality in the repository is

$$\|A - X\|_F^2 < (d-1)E_* \qquad (E_* > 0), \tag{1}$$

with exact reconstruction when $E_* = 0$, in the idealized arithmetic/SVD model [3, 2]. A guarantee of the form (1) for every input does not imply that the supremum of all error ratios is smaller than $d-1$. We prove the former, and make no claim of the latter. In particular this note must not be presented as a smaller uniform worst-case constant.

*Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Cambridge, United Kingdom. Email: m.colbrook@damtp.cam.ac.uk.

2 A completion estimate

The standard TT-SVD guarantee [1, Theorem 2.2 and Corollary 2.4] will be used on the remaining cuts after a first projection.

Lemma 1. *Let $U \in \mathbb{R}^{n_1 \times k}$ have orthonormal columns, with $k \leq r_1$, and put $P = UU^T$. Compress the first mode to form $B = U^T A$. Treat its first two indices as a single index, so B is a tensor of order $d-1$ and format $(kn_2) \times n_3 \times \cdots \times n_d$. Apply ordinary TT-SVD to this tensor with cut-rank bounds $r_2, \ldots, r_{d-1}$, and lift the result through U to obtain X_U. Then $X_U \in S_r$, and for every $Y \in S_r$,*

$$\|A - X_U\|_F^2 \leq \|(I-P)A\|_F^2 + (d-2)\|P(A-Y)\|_F^2. \tag{2}$$

Here and below a matrix acting on a tensor acts on its first mode.

Proof. The first cut of X_U has rank at most k. The remaining cuts have the specified rank bounds, which are preserved by lifting through U. The compressed tensor $U^T Y$ satisfies every remaining cut constraint: compression is left multiplication on each such unfolding and cannot increase its rank. The TT-SVD guarantee on the $d-2$ remaining cuts is therefore

$$\|B - \widehat{B}\|_F^2 \leq (d-2)\|U^T(A-Y)\|_F^2.$$

For completeness, this usual guarantee follows by summing the squared orthogonal truncation residuals at successive cuts. Each residual is at most the best error at that cut; projecting any feasible comparison tensor through the previously computed orthogonal factors shows that each is at most the squared error of that comparison tensor. There are $d-2$ cuts. This argument also gives zero residuals when the input is already feasible. Finally, the first-mode discarded component is orthogonal to the lifted completion residual, giving

$$\|A - X_U\|_F^2 = \|(I-P)A\|_F^2 + \|B - \widehat{B}\|_F^2.$$

The identity $\|U^T(A-Y)\|_F = \|P(A-Y)\|_F$ completes the proof. $\square$

3 The finite algorithm

Let $A_{(1)}$ denote the $n_1 \times (n_2 \cdots n_d)$ first unfolding. If $A = 0$, return 0. Compute its SVD and let

$$\varepsilon^2 = \sum_{i > r_1} \sigma_i(A_{(1)})^2,$$

where singular values beyond the matrix dimensions are interpreted as zero. Construct a list $\mathcal{P}$ of first-mode projectors as follows.

If $\operatorname{rank} A_{(1)} \leq r_1$, use just the projector onto the column space of $A_{(1)}$. Otherwise let $\tau = \sigma_{r_1}(A_{(1)}) > 0$. Let H be the span of left singular vectors with singular value strictly greater than τ, put $h = \dim H$, and let E be the left singular space for τ, of dimension t. Set $s = r_1 - h$, so $1 \leq s \leq t$. If $s = t$, use just the projector onto $H \oplus E$. If $s < t$, take any fixed orthonormal basis $u_0, \ldots, u_{t-1}$ of E and form the t cyclic windows

$$W_j = \operatorname{span}\{u_j, u_{j+1}, \ldots, u_{j+s-1}\}, \qquad 0 \leq j < t,$$

with subscripts modulo t. Use the t projectors

$$P_j = P_H + P_{W_j}. \tag{3}$$

Each projector in $\mathcal{P}$ is a best first-cut rank-r_1 projection (or an exact column-space projection when the rank is smaller). For each $P \in \mathcal{P}$, choose an orthonormal column matrix U with

$P = UU^T$, form X_U by Lemma 1, and compute its actual Frobenius error. Return a candidate of minimum error, resolving any final tie deterministically.

There are at most n_1 candidates. SVDs, matrix products, tensor reshapings, and error computations have polynomial cost in the dense input size and the rank parameters. Thus this is a deterministic polynomial-time algorithm in the arithmetic/SVD model. Equality of singular values is tested exactly in that model. A floating-point implementation with a tolerance is an illustration, not a bit-complexity implementation of the exact tie test.

4 Proof of the strict guarantee

We first isolate the only equality case requiring attention.

Lemma 2. *Suppose* $\operatorname{rank} A_{(1)} > r_1$, *and let* Y *satisfy* $\operatorname{rank} Y_{(1)} \le r_1$ *and* $\|A - Y\|_F^2 = \varepsilon^2$. *With* H *and* E *defined above, the residual* $R = A - Y$ *obeys* $P_H R = 0$. *If* $s = t$, *it also obeys* $(P_H + P_E)R = 0$.

Proof. Let S be the column space of $Y_{(1)}$. The orthogonal decomposition

$$\|A_{(1)} - Y_{(1)}\|_F^2 = \|(I - P_S)A_{(1)}\|_F^2 + \|P_S A_{(1)} - Y_{(1)}\|_F^2$$

shows that equality with the best rank-r_1 error forces $Y_{(1)} = P_S A_{(1)}$ and forces P_S to maximize the captured singular energy. Since $\tau > 0$, $\dim S = r_1$. The equality characterization of this variational maximum gives $S = H \oplus E_0$ for an s-dimensional subspace $E_0 \subset E$. One can see this directly by writing $\operatorname{tr}(P_S A_{(1)} A_{(1)}^T)$ in an eigenbasis: all weights at eigenvalues strictly above τ^2 must be one, all below must be zero, and the remaining weight is s in the τ^2 eigenspace. Thus $H \subset S$, and also $E \subset S$ when $s = t$. The asserted residual orthogonality follows. ☐

Theorem 3. *The finite algorithm above returns* $X \in S_r$ *satisfying* (1) *whenever* $E_* > 0$, *and returns* $X = A$ *when* $E_* = 0$.

Proof. Fix a best feasible $Y \in S_r$, and write $R = A - Y$. The first-cut lower bound is $\varepsilon^2 \le E_*$. Every tested projector P has $\|(I - P)A\|_F^2 = \varepsilon^2$; Lemma 1 gives

$$\|A - X_U\|_F^2 \le \varepsilon^2 + (d-2)\|PR\|_F^2 \le \varepsilon^2 + (d-2)E_*. \tag{4}$$

If $\varepsilon^2 < E_*$, this is strictly smaller than $(d-1)E_*$. In particular, this covers $\operatorname{rank} A_{(1)} \le r_1$ whenever $E_* > 0$.

It remains to consider $\varepsilon^2 = E_* > 0$. By Lemma 2, $P_H R = 0$. If $s = t$, the single tested projector annihilates R, so (4) bounds the error by $E_* < (d-1)E_*$. If $0 < s < t$, the cyclic-window projectors satisfy the exact averaging identity

$$\frac{1}{t}\sum_{j=0}^{t-1} P_{W_j} = \frac{s}{t} P_E,$$

since each basis vector appears in exactly s windows. Hence

$$\frac{1}{t}\sum_{j=0}^{t-1} \|P_j R\|_F^2 = \frac{s}{t}\|P_E R\|_F^2 \le \frac{s}{t} E_* < E_*.$$

At least one tested projector has this bound. Its completion satisfies

$$\|A - X_U\|_F^2 \le \left(1 + (d-2)\frac{s}{t}\right) E_* < (d-1)E_*.$$

The returned minimum-error candidate is no worse.

Finally, if $E_* = 0$, then $A \in S_r$, $\varepsilon^2 = 0$, and (4) with $R = 0$ forces the returned error to be zero. ☐

5 Why this is not a uniform constant improvement

This distinction is material to the scope of the result, not a numerical precision issue. Consider $d = 3$, format $3 \times 3 \times 2$, ranks $(2, 1)$, and the tensor

$$A = \alpha e_1 \otimes e_1 \otimes f_1 + \beta e_2 \otimes e_2 \otimes f_1 + \gamma e_3 \otimes e_3 \otimes f_2, \qquad \alpha > \gamma \geq \beta > 0.$$

Its optimal error is $E_* = \gamma^2$. Indeed, a tensor feasible for ranks $(2, 1)$ is $M \otimes v$ with rank $M \leq 2$. For a unit vector $v = (c, s)$, the contracted matrix is $\operatorname{diag}(\alpha c, \beta c, \gamma s)$. The maximum possible sum of its two largest squared singular values, also maximizing over $c^2 + s^2 = 1$, is

$$\max\{\alpha^2 + \beta^2, \alpha^2, \beta^2, \gamma^2\} = \alpha^2 + \beta^2.$$

This proves the claimed optimum by orthogonal projection.

When $\gamma > \beta$, the best first-cut rank-two subspace is uniquely $\operatorname{span}(e_1, e_3)$. The remaining rank-one truncation retains only the α term. The algorithm's squared error is therefore $\beta^2 + \gamma^2$, and its ratio to optimum is

$$1 + (\beta/\gamma)^2 < 2.$$

Keeping $\alpha = 2$, $\beta = 1$ and sending $\gamma \downarrow 1$ shows that these ratios approach $2 = d - 1$ in a fixed format. At $\gamma = \beta$, an unfortunate ordinary TT-SVD tie choice can attain ratio exactly two; the finite tie handling removes that equality. No theorem here asserts a uniform ratio bounded by $2 - \delta$ for a fixed $\delta > 0$.

6 Reproducibility and scope

The accompanying `verify_tr04.py` checks the projector averaging identity exactly, checks the fixed-format family and rotated tie bases, and tests the numerical implementation for rank feasibility and exact reconstruction up to floating-point error. The exact mathematical algorithm and its guarantee are Theorem 3; a tolerance-based SVD is not claimed to detect exact ties in finite precision. No ranks are enlarged. No oracle for E_* or for an optimal Y is used: those objects appear only in the proof. The assertion addressed is the strict pointwise target, not a stronger uniform-factor or bit-complexity formulation.

References

[1] I. V. Oseledets, *Tensor-Train Decomposition*, SIAM Journal on Scientific Computing 33 (2011), 2295–2317, Theorem 2.2 and Corollary 2.4. doi:10.1137/090752286.

[2] N. Amsel et al., *Linear Systems and Eigenvalue Problems: Open Questions from a Simons Workshop*, arXiv:2602.05394v3 (2026), Section 6.1, Problem 6.1.

[3] OpenProblemsInNLA, *TR-04: Improve the worst-case approximation factor for prescribed tensor-train ranks*, canonical statement. `https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/tensor-computations/TR-04`.